ECONOMICS

Q1-31 Memory Notes | Exam-ready answers for fast revision

RED: MUST MEMORIZE	YELLOW: UNDERSTAND	GREEN: FORMULA	BLUE: EXAMPLE
Definitions / key laws	Concept / explanation	Write exactly in exam	Use for easy marks

How to use tonight: Read the 30-second line first. Then say the mnemonic aloud. Finally, cover the answer and reproduce the bold keywords.

1. Meaning of Economics

EXAM-READY ANSWER

Economics is the social science that studies how people, firms and governments use **scarce resources** to satisfy **unlimited wants**. Since resources have alternative uses, people must make choices.

MEMORY BOX
Scarcity -> Choice -> Allocation

30-SECOND REVISION: Economics = choice under scarcity.

2. Definitions of Economics

EXAM-READY ANSWER

Adam Smith: study of wealth of nations. **Marshall:** study of mankind in ordinary business life and material welfare. **Robbins:** human behaviour between unlimited ends and scarce means with alternative uses. **Samuelson:** how society uses scarce resources to produce and distribute goods over time, including growth.

Comparison: Wealth -> Welfare -> Scarcity/Choice -> Growth and Distribution.

MEMORY BOX
S M R S = Wealth, Welfare, Scarcity, Society grows.

30-SECOND REVISION: Write the four names in order: Smith, Marshall, Robbins, Samuelson.

3. Importance of studying Economics

EXAM-READY ANSWER

Economics helps: (1) individuals spend, save and invest wisely; (2) firms decide what and how much to produce; (3) government make policy on tax, jobs and inflation; (4) society understand poverty, unemployment and inequality; and (5) countries understand trade and growth.

MEMORY BOX
I-F-G-S-T: Individual, Firm, Government, Society, Trade.

30-SECOND REVISION: It teaches the wise use of limited resources at every level.

4. Scientific methods in microeconomics

EXAM-READY ANSWER

Deductive method: starts with a general principle or assumption and reaches a specific conclusion. Example: price falls, so quantity demanded rises.

Inductive method: starts with observations/data and develops a general law.

Modern economics uses both, with models, statistics and *ceteris paribus* (other things constant).

MEMORY BOX
D = Down: general rule down to case. I = Increase: cases build up to rule.

30-SECOND REVISION: Deduction: general to particular. Induction: particular to general.

5. Importance of economic models

EXAM-READY ANSWER

An economic model is a **simplified representation of reality** based on assumptions. It simplifies complex problems, shows relationships, predicts results, tests policies before use, and guides business decisions.

Example: demand-supply model predicts that a fall in rice supply raises its price.

MEMORY BOX
S-P-T-G: Simplify, Predict, Test, Guide.

30-SECOND REVISION: A model is not real life; it is a useful simplified map of real life.

6. Taxes and subsidies: consumer and producer surplus

EXAM-READY ANSWER

Tax: consumers pay a higher price; producers receive a lower net price. Both consumer and producer surplus fall. Government receives tax revenue, but some trades disappear, causing **deadweight loss**.

Subsidy: consumers pay less and producers receive more; both surpluses rise. But government pays the subsidy and overproduction/overconsumption can create deadweight loss.

MEMORY BOX
Tax = Take surplus. Subsidy = Support surplus.

30-SECOND REVISION: Tax creates a wedge: buyer pays more, seller gets less.

7. Interest rates and inter-temporal consumption

EXAM-READY ANSWER

Inter-temporal consumption means choosing consumption **now** versus **later**. A higher interest rate rewards saving, so it usually encourages saving and future consumption. It also makes borrowing costly, so borrowers reduce present consumption. A lower rate makes saving less attractive and borrowing cheaper, so present consumption tends to rise.

MEMORY BOX
High rate: Save. Low rate: Spend.

30-SECOND REVISION: Interest is the price of shifting consumption through time.

8. Role of interest rate in inter-temporal choice

EXAM-READY ANSWER

A consumer chooses present consumption (C1) and future consumption (C2). The budget-line slope is **-(1+r)**. Interest rate determines the relative price of present and future consumption. When r rises, the line becomes steeper and saving is more rewarding. At optimum: **MRS = 1 + r**.

MEMORY BOX
r rises -> slope rises -> saving rewards rise.

30-SECOND REVISION: Key condition: MRS = 1 + r.

9. Short-run equilibrium: perfect competition

EXAM-READY ANSWER

A competitive firm is a **price taker**, so **AR = MR = P**. It maximizes profit where **MC = MR** and MC cuts MR from below.

If P > AC: supernormal profit. If P = AC: normal profit. If AVC < P < AC: loss but continue in short run. If P < AVC: shut down.

MEMORY BOX
MC = MR; P vs AC tells profit; P vs AVC tells shutdown.

30-SECOND REVISION: Draw horizontal AR=MR=P; MC cuts it from below.

10. Monopoly: short-run and long-run equilibrium

EXAM-READY ANSWER

A monopolist is a single seller. Its demand/AR curve slopes downward and **MR lies below AR**. It produces where **MC = MR**, then takes price from the AR curve.

In the short run it may earn profit, normal profit or loss. In the long run, entry barriers prevent competitors, so supernormal profit can continue.

MEMORY BOX
Monopoly: MR below AR; barriers protect profit.

30-SECOND REVISION: Unlike perfect competition, monopoly may keep abnormal profit in the long run.

11. Income and price changes on budget line

EXAM-READY ANSWER

Budget equation: **M = PxX + PyY**.

Income changes: income rises -> parallel outward shift; income falls -> parallel inward shift. Slope stays unchanged.

Price of X changes: Px falls -> X-intercept moves out and line becomes flatter; Px rises -> X-intercept moves in and line becomes steeper. A proportional change in both prices acts like an opposite income change.

MEMORY BOX
Income = Shift; one price = Rotate.

30-SECOND REVISION: Remember: income shifts, price pivots.

12. Expected utility under uncertainty

EXAM-READY ANSWER

Expected utility is the probability-weighted average of utility: **EU = p1U(X1) + p2U(X2) + ...**. A rational person chooses the option with the highest expected utility, not necessarily the highest expected money value. Risk-averse people often prefer a certain amount because marginal utility of wealth diminishes.

MEMORY BOX
P-U: Probability x Utility, then add.

30-SECOND REVISION: Expected utility uses utility, not only money.

13. Microeconomics vs Macroeconomics

EXAM-READY ANSWER

Microeconomics: individual units - consumer, firm, industry; demand, supply, price and cost. It is price theory.

Macroeconomics: whole economy - national income, employment, inflation and growth. It is income theory.

MEMORY BOX
Micro = microscope = small units. Macro = map = whole economy.

30-SECOND REVISION: Micro studies parts; macro studies the whole.

14. Economic goods vs free goods

EXAM-READY ANSWER

Economic goods are scarce relative to wants, have price/cost and opportunity cost. Examples: food, cars.

Free goods are abundant relative to wants, have no price and no opportunity cost. Examples: sunlight and air in normal conditions.

The key difference is **scarcity**.

MEMORY BOX
Economic = scarce = sacrifice. Free = abundant.

30-SECOND REVISION: If it is scarce, it is an economic good.

15. Central problems of an economy

EXAM-READY ANSWER

Because resources are scarce, every economy decides:
1. What to produce and how much?
2. How to produce? (labour-intensive or capital-intensive)
3. For whom to produce? (distribution)

Modern economics also asks how to achieve growth.

MEMORY BOX
W-H-F-G: What, How, For whom, Growth.

30-SECOND REVISION: Scarcity creates the three questions: What, How, For whom?

16. Needs, wants and demand

EXAM-READY ANSWER

Need: basic necessity for survival, e.g., food. **Want:** a specific desire, e.g., pizza. **Demand:** a want backed by **ability and willingness to pay** at a given price.

Thus: Need -> Want -> Demand. Not every want becomes demand.

MEMORY BOX
Demand = Want + Money + Willingness.

30-SECOND REVISION: A desire without purchasing power is a want, not demand.

17. Total utility and marginal utility

EXAM-READY ANSWER

TU is total satisfaction from all units. **MU** is extra satisfaction from one more unit: **MU = Delta TU / Delta Q**.

MU positive -> TU rises. MU = 0 -> TU is maximum. MU negative -> TU falls. MU is the slope of the TU curve.

MEMORY BOX
Positive MU: TU up. Zero MU: TU top. Negative MU: TU down.

30-SECOND REVISION: The turning point: MU = 0, TU maximum.

18. Opportunity cost

EXAM-READY ANSWER

Opportunity cost is the **value of the next best alternative forgone** when a choice is made. It is the real economic cost, not just money paid.

Example: If 3 hours of study replace a job earning $30, the opportunity cost of studying is $30.

MEMORY BOX
Choice costs the next best chance.

30-SECOND REVISION: Opportunity cost = next best alternative sacrificed.

19. Demand

EXAM-READY ANSWER

Demand is the quantity of a good a consumer is **willing and able** to buy at a **given price** during a **given period**, other things constant. It is not merely a desire.

MEMORY BOX
W-A-P-T: Willing, Able, Price, Time.

30-SECOND REVISION: Demand always needs willingness, ability, price and time.

20. Law of Demand

EXAM-READY ANSWER

Other things remaining constant, price and quantity demanded have an inverse relationship: price rises -> quantity demanded falls; price falls -> quantity demanded rises. Therefore the demand curve slopes downward. Reasons: **substitution effect** and **income effect**.

MEMORY BOX
Price up, demand down; Price down, demand up.

30-SECOND REVISION: Use the phrase ceteris paribus for full marks.

21. Exceptions to Law of Demand

EXAM-READY ANSWER

Main exceptions are:
Giffen goods - poor households buy more of a staple when its price rises.
Veblen/prestige goods - high price adds status.
Speculation - buyers expect further price rises.
Necessities - life-saving goods have little response.
Ignorance/emergency - unusual buying behaviour.

MEMORY BOX
GVSNI: Giffen, Veblen, Speculation, Necessity, Ignorance.

30-SECOND REVISION: Five exceptions: GVSNI.

22. Production

EXAM-READY ANSWER

Production is the process of combining inputs - land, labour, capital and entrepreneurship - to create goods and services that satisfy wants. It includes creating **form, place, time and possession utility**; transport is production because it adds place utility.

MEMORY BOX
LLCE makes utility: Land, Labour, Capital, Entrepreneur.

30-SECOND REVISION: Production means creating utility, not only making factory goods.

23. Factors of production

EXAM-READY ANSWER

Land: natural resources; reward = **rent**.
Labour: human physical and mental effort; reward = **wages**.
Capital: man-made tools/machines/buildings; reward = **interest**.
Entrepreneur: organizes factors and bears risk; reward = **profit**.

MEMORY BOX
LLCE -> RWIP: Land-Rent, Labour-Wages, Capital-Interest, Entrepreneur-Profit.

30-SECOND REVISION: Memorize the pair chain: Land-Rent, Labour-Wages, Capital-Interest, Entrepreneur-Profit.

24. Law of Diminishing Marginal Returns

EXAM-READY ANSWER

When more units of a **variable factor** are added to a **fixed factor**, with technology constant, the extra output from each additional unit eventually falls.

Example: Adding more workers to one acre of land eventually causes crowding, so each new worker adds less output.

MEMORY BOX
More workers + same land = less extra output.

30-SECOND REVISION: Use the words variable factor, fixed factor, eventually falls.

25. Marginal Rate of Technical Substitution (MRTS)

EXAM-READY ANSWER

MRTS is the rate at which one input can replace another while output stays constant on the same isoquant. **MRTS(L for K) = -Delta K/Delta L = MPL/MPK.** Usually MRTS diminishes as more labour replaces capital; therefore isoquants are convex to the origin.

MEMORY BOX
Same output, swap inputs.

30-SECOND REVISION: MRTS = labour for capital without changing output.

26. Cobb-Douglas production function

EXAM-READY ANSWER

Q = A L^a K^b, where Q = output, A = technology, L = labour, K = capital, and a/b are output elasticities.

Returns to scale: **a+b=1** constant; **a+b>1** increasing; **a+b<1** decreasing. It also shows diminishing marginal product of each input when the other is fixed.

MEMORY BOX
ALK: A = technology, L = labour, K = capital. Sum a+b tells scale.

30-SECOND REVISION: For returns to scale, only check a + b.

27. Production Possibility Curve (PPC)

EXAM-READY ANSWER

A PPC shows the maximum combinations of two goods an economy can produce by using all resources and technology efficiently.

Points **on** the curve are efficient; **inside** are inefficient/unemployed resources; **outside** are unattainable now. Its slope shows opportunity cost. It is normally bowed outward because opportunity cost increases.

MEMORY BOX
O-I-O: On = efficient, Inside = inefficient, Outside = impossible.

30-SECOND REVISION: PPC teaches scarcity, choice and opportunity cost.

28. Derivation of individual demand curve

EXAM-READY ANSWER

Marginal utility approach: consumer buys until price equals marginal utility (in money). Since MU falls as quantity rises, a lower price leads to more purchase, giving a downward demand curve.

Indifference-curve approach: change Px while income and Py stay constant; each new budget-line tangency gives an optimum quantity. Plot price against those quantities to obtain demand.

MEMORY BOX
MU falls -> Price must fall -> Quantity rises.

30-SECOND REVISION: Demand curve comes from different optimal quantities at different prices.

29. Market equilibrium

EXAM-READY ANSWER

Market equilibrium occurs where **quantity demanded = quantity supplied**. This gives equilibrium price and quantity. It is the intersection of demand and supply curves. Above equilibrium price there is surplus, which pushes price down. Below equilibrium price there is shortage, which pushes price up.

MEMORY BOX
High price = surplus = down. Low price = shortage = up.

30-SECOND REVISION: Equilibrium means Qd = Qs.

30. Equilibrium price and output through demand and supply

EXAM-READY ANSWER

Equilibrium is determined at the intersection of market demand and supply.

At price above equilibrium: **Qs > Qd**, surplus arises, sellers cut price.
At price below equilibrium: **Qd > Qs**, shortage arises, buyers bid price up.

Adjustment continues until **Qd = Qs**, giving **P*** and **Q***.

MEMORY BOX
Above: Supply wins. Below: Demand wins.

30-SECOND REVISION: Write both adjustment paths and finish with Qd = Qs.

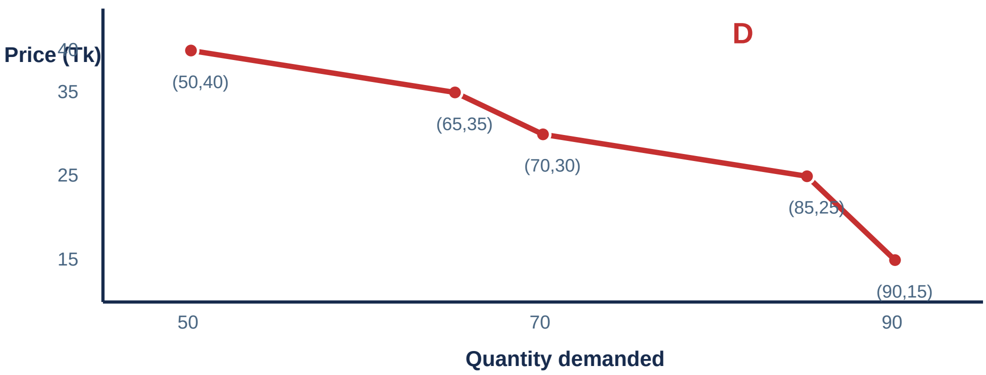
31. Demand curve from the data

EXAM-READY ANSWER

Data: Price (Tk) = 15, 25, 30, 35, 40; Quantity demanded = 90, 85, 70, 65, 50.

Put **Quantity** on X-axis and **Price** on Y-axis. Plot points: (Q,P) = (90,15), (85,25), (70,30), (65,35), (50,40). Join them with a smooth downward-sloping line from upper left to lower right.

Conclusion: as price rises from 15 to 40, quantity demanded falls from 90 to 50; therefore it follows the Law of Demand.



MEMORY BOX
Q on X, P on Y. Price up -> quantity down.

30-SECOND REVISION: For the graph: X = quantity; Y = price; downward slope.